

Finite-Dimensional Approximations and Numerical Analysis of the Parallel Key Geometric Flow:

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Abstract

This paper develops finite-dimensional approximations of the Parallel Key Geometric Flow (PKGF) and analyzes the resulting numerical schemes based on Galerkin discretization. Through these approximations, we investigate the discrete behavior of the **conservative–dissipative–unified (C/D/U)** dynamics that characterize the unified phase of PKGF.

While PKGF is formulated as an operator flow on an infinite-dimensional Hilbert bundle, numerical experiments and applications—such as SLAM, OCR, structural abstraction, and computational models in Noetics—require finite-dimensional realizations. This work numerically reinforces the theoretical framework that views intelligence as a physical process governed by the balance of conservation and dissipation [Fagan 2026].

The goals of this paper are:

1. to construct Galerkin approximations for both linear and nonlinear PKGF,
2. to establish convergence and stability of the discrete solutions,
3. to analyze the preservation and decay of discrete energy structures, and
4. to numerically reproduce spectral flow and rank transitions in the unified phase.

The results provide a bridge between the infinite-dimensional theory of PKGF and its computational implementations.

Keywords: Parallel Key Geometric Flow, PKGF, Galerkin Discretization, Numerical Analysis, Unified-Phase Dynamics, Finite-Dimensional Approximation

1 Introduction

The Parallel Key Geometric Flow (PKGF) is an operator evolution equation that unifies a conservative commutator term with a dissipative elliptic term. It forms a mathematical foundation for Noetics and operator-level structural dynamics [Fagan 2026].

Previous work has established:

- well-posedness of the linear PKGF,
- spectral flow behavior in the unified phase,
- comparison of PKGFs energy structure with GENERIC and Onsager frameworks [Chen et al. 2024],
- and well-posedness of nonlinear PKGF.

However, PKGF is inherently infinite-dimensional, and **finite-dimensional approximations are essential** for numerical experiments and applications.

This paper develops Galerkin approximations of PKGF and analyzes their convergence, stability, and discrete energy behavior.

2 Construction of Galerkin Approximations

Let

$$H = L^2(M, \text{End}(E))$$

be the Hilbert space of bundle endomorphisms. We choose finite-dimensional subspaces

$$H_1 \subset H_2 \subset \cdots \subset H,$$

and denote by

$$P_n : H \rightarrow H_n$$

the **orthogonal projection** onto H_n . In our discretization, we apply insights from high-order discontinuous Galerkin spectral element methods (DGSEM) to ensure physical and entropy stability [Coquel et al. 2021].

2.1 Choice of Basis

The construction of H_n may use:

- eigenfunctions of the Laplace operator Δ ,
- finite element bases reflecting the geometry of the substrate,
- or structural bases satisfying Noetics principle of **substrate invariance**.

This ensures that the geometric properties of PKGF are preserved under discretization.

2.2 Discretization of Nonlinear Terms

Since the nonlinear potential $\Omega(K)$ is an operator-valued function on H , the **connection itself must be discretized**.

We define:

$$\Omega_n(K_n) = P_n \Omega(K_n) P_n,$$

so that $\Omega_n(K_n)$ remains in H_n . This aligns the theory with matrix-based numerical implementations.

3 Discrete PKGF Equations

For the linear PKGF:

$$\partial_t K = [\Omega, K] + \lambda \mathcal{D}(K),$$

the Galerkin approximation is:

$$\partial_t K_n = P_n([\Omega, K_n]) + \lambda P_n(\mathcal{D}(K_n)).$$

For the nonlinear PKGF:

$$\partial_t K = [\Omega(K), K] + \lambda \mathcal{D}(K) + \mathcal{N}(K),$$

the discrete system becomes:

$$\partial_t K_n = P_n([\Omega_n(K_n), K_n]) + \lambda P_n(\mathcal{D}(K_n)) + P_n(\mathcal{N}(K_n)).$$

4 Convergence Analysis

4.1 Interaction Between the Conservative Term and Projection

Because P_n is an **orthogonal projection** and $K_n \in H_n$, we have:

$$\langle P_n([\Omega, K_n]), K_n \rangle = \langle [\Omega, K_n], K_n \rangle.$$

The skew-adjointness of the commutator operator implies:

$$\langle [\Omega, K_n], K_n \rangle = 0.$$

Thus, the conservative term does not contribute to the discrete energy, just as in the continuous setting.

4.2 Convergence of Mild Solutions

Let $K(t)$ be the solution of the linear PKGF. Then the Galerkin approximations satisfy:

$$\sup_{t \in [0, T]} \|K_n(t) - K(t)\|_{L^2} \rightarrow 0.$$

The proof utilizes analytic techniques related to Łojasiewicz–Simon gradient inequalities, which govern the asymptotic convergence of gradient flows [Feehan and Maridakis 2019].

4.3 Convergence for Nonlinear PKGF

If $\mathcal{N}$ is locally Lipschitz, then:

$$K_n \rightarrow K \quad \text{in } C([0, T]; L^2).$$

5 Discrete Energy Preservation and Decay

Define the discrete energy:

$$E_n(K_n) = \frac{1}{2} \|K_n\|_{L^2}^2.$$

5.1 Theorem 5.1 (Discrete Energy Decay)

$$\frac{d}{dt} E_n(K_n(t)) = \langle P_n(\mathcal{D}(K_n)), K_n \rangle \leq 0.$$

The conservative term does not contribute to the energy. This structure is highly compatible with energy-stable and structure-preserving numerical schemes based on the Onsager principle [Chen et al. 2024].

6 Numerical Reproduction of Unified-Phase Dynamics

The unified phase involves the complexified operator:

$$\tilde{K} = K_{\text{core}} + iK_{\text{fluct}},$$

which induces rotations, collapses, and crossings of eigenvalues.

6.1 Discrete Spectral Flow

For the discrete operator:

$$\mathcal{L}_n(t) = \mathcal{L}_0 + K_n(t),$$

we track eigenvalues $\lambda_{n,j}(t)$ to compute:

$$\text{SF}(\mathcal{L}_n(t)).$$

K-theoretic computational methods for lattice-based operators [Aoki et al. 2025] provide critical insights into the formulation of spectral flow in discrete spaces.

6.2 Numerical Robustness: -thresholds and Avoided Crossings

In numerical computations, exact zero-crossings are difficult to detect. We therefore use:

- sign changes of eigenvalues,
- -thresholds for rank determination,
- and distinction between **avoided crossings** and true crossings.

Analytic results on the stability of Fredholm operators and gap topology [Waterstraat 2016] ensure the numerical robustness of the discrete spectral flow.

7 Numerical Experiments (Conceptual Examples)

1. Laplace-type PKGF on the 1D torus
2. Matrix-valued PKGF (finite-dimensional model)
3. Nonlinear abstraction model $\mathcal{N}(K) = -\alpha K^2$
4. **Visualization of rank transitions in the unified phase**

8 Conclusion and Future Work

This paper developed finite-dimensional approximations of PKGF and established:

- convergence of Galerkin approximations,
- preservation and decay of discrete energy,
- numerical reproduction of spectral flow and rank transitions.

Future work includes:

- **geometric time integration** for mixed conservative–dissipative flows,
- applying the LaSalle invariance principle for discrete-time stability analysis [Mei and Bullo 2020],
- numerical analysis of nonlinear spectral flow,
- relationships between discretization error and topological invariants,
- computational models of concept formation in Noetics.

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